AGRIVIVE / PHAD-FLOW

MAHARASHTRA / PHAD-INSPIRED / ROTARY WATER ALLOCATION VALVE

SERIES 14 / 28

A3 / PRELIMINARY / ALL DIMENSIONS mm



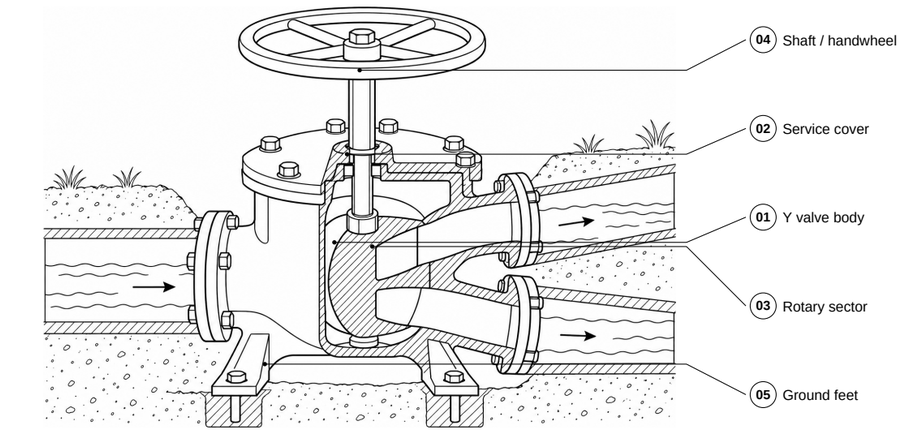
STATE CONTEXT / INDIGENOUS INSPIRATION

Phad irrigation in Maharashtra uses organised diversion and distribution of river water. PHAD-FLOW adapts the allocation idea into a manually operated three-port valve; branch delivery requires calibration.

Practice source: CGWB manual: traditional Phad irrigation | Outline: geoBoundaries / schematic

01 INSTALLED SOIL / WATER SECTION

NOMINAL ENVELOPE / mm: 160 L x 100 W x 100 H

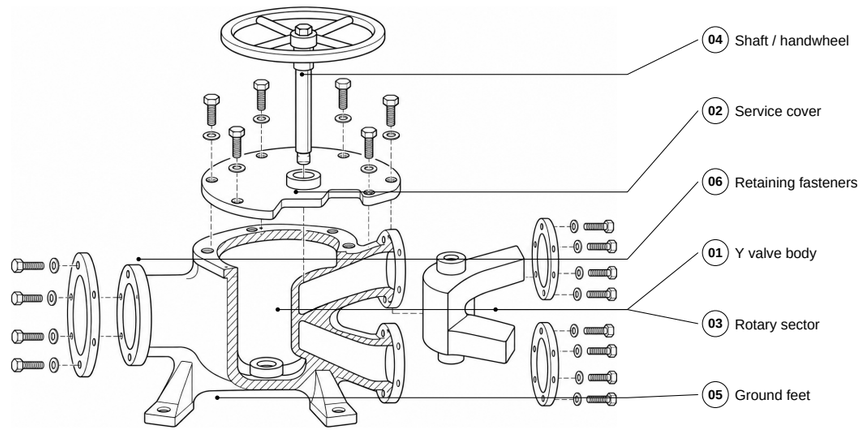


DESIGN / OPERATION NOTES

Anchor the body at a suitable gravity junction.
Fit the rotary sector and service cover, then operate the handwheel.
Sector position can select a branch; flow is not inferred from wheel angle alone.

02 EXPLODED ASSEMBLY

ASSEMBLED ENVELOPE / mm
160 L x 100 W x 100 H

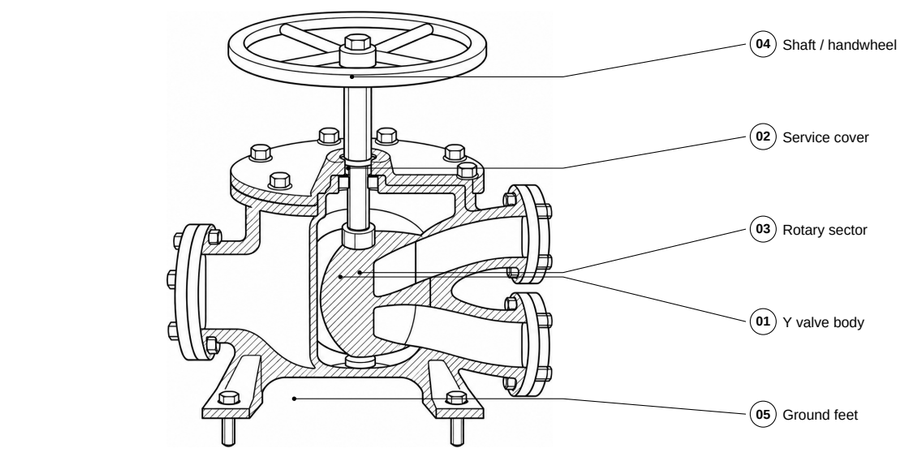


DESIGN / OPERATION NOTES

Components separated for service and assembly.
Connections and mating fits require CAD.

04 INTERNAL SECTION

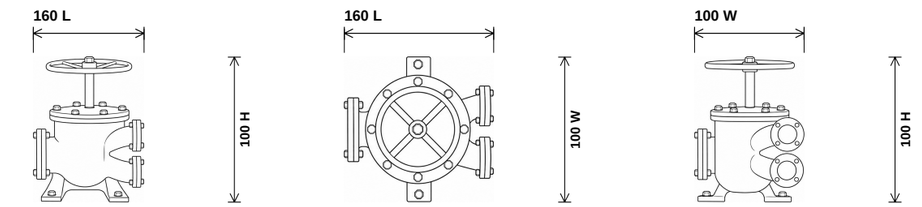
SECTION BASIS / mm
25 inlet ID; two 20 outlet IDs; rotary gate clearances TBD



DESIGN / OPERATION NOTES

Fit the rotary sector and service cover, then operate the handwheel.
Sector position can select a branch; flow is not inferred from wheel angle alone.

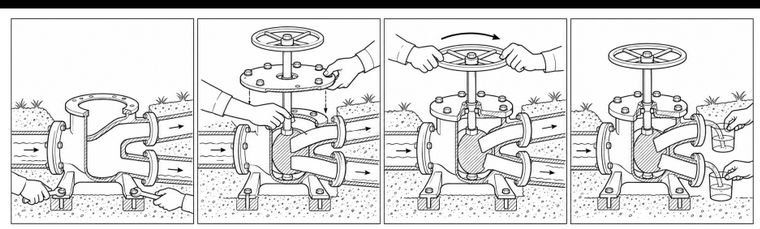
03 ORTHOGRAPHIC STUDIES / mm



FRONT **TOP** **SIDE**

Body 160 L x 100 W x 100 H. Inlet DIA 25 ID; two outlets DIA 20 ID; rotary clearances TBD.

05 INSTALLATION / OPERATING SEQUENCE



1 ANCHOR BODY **2 FIT ROTOR** **3 SELECT BRANCH** **4 CALIBRATE FLOW**

Prototype fit, access and operation require trials.

DESIGN / INTERFACE SCHEDULE

01 NOMINAL GEOMETRY

External L x W x H
160 x 100 x 100

Inlet ID
25

Two outlet IDs
20 each

02 GEOMETRIC CHECK

COMBINED OUTLET BORE AREA

628.3 mm2

2 x pi/4 x 20^2
Allocation depends on head and calibrated gate position

03 PARTS / QTY

01 Y valve body	**1**
02 Service cover	**1**
03 Rotary sector	**1**
04 Shaft / handwheel	**1 set**
05 Ground feet	**Integral**
06 Retaining fasteners	**1 set**

Body/rotor/cover: candidate polymer. Shaft, fasteners and seals: compatible grades TBD.

04 INTERFACES / CHECKS

Test sector sealing, friction and debris clearance.

Measure branch flow at representative heads.

Check shaft retention and cover access.

PRELIMINARY / VALIDATION PENDING

DESIGN DEVELOPMENT RECORD

01 / GEOMETRY
Nominal body sizes are proposed. Generated views are illustrative; small features and interfaces may vary.

02 / BEFORE FABRICATION
Develop one CAD model, assign wall thicknesses, tolerances and mating fits. Check print orientation and service access.

03 / TRIAL RECORD
Measure dimensions and retained volume or opening area. Record leakage, clogging and soil anchorage under observed conditions.

No discharge, treatment or durability performance is established.

DRAWING / REVISION	TITLE	STATUS	REVISION DATE	FORMAT / SCALE / UNITS	COLLECTION SHEET
AGR-14 / P01	PHAD-FLOW	PRELIMINARY CONCEPT / CAD & TRIALS PENDING	10 NOV 2026	A3 / NTS / mm	14 / 28